

Literature Benchmark Brief: Li_2025_ApplEnergy

Title: A novel frequency-domain physics-informed neural network for accurate prediction of 3D spatio-temporal wind fields

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Cluster: Cluster A: Spatio-Temporal Wind/Wake

Core Theoretical Mechanism & Model Formulation

Frequency-domain PINN decomposition for turbulent velocity fluctuations

Empirical Scale & Experimental Dataset

3D complex terrain wind turbine array

Relevance to IEEE TSTE & Sustainable Energy Systems

Highlights spectral degradation of wind turbulence across spatial scales

Identified Gaps & Operational Vulnerabilities

High computational complexity in frequency transformation; impractical for edge devices

Role in Manuscript Defense & Reviewer #2 Proof

Reinforces why compact time-domain directed GRU is selected for substation deployment